

Advances with antibody-drug conjugates in breast cancer treatment.

PMID: 34748933 | Indexed in PubMed

Antibody-drug conjugate-based therapy for treatment of cancer has attracted much attention because of its enhanced efficacy against numerous cancer types. Commonly, an ADC includes a mAb linked to a therapeutic payload. Antibody, linker and payload are the three main components of ADCs. The high specificity of antibodies is integrated with the strong potency of payloads in ADCs. ADCs with potential cytotoxic small molecules as payloads, generate antibody-mediated cancer therapy. Recently, ADCs with DNA-damaging agents have shown favor over microtubule-targeting agents as payloads. Although ADC resistance can be a barrier to effectiveness, several ADC therapies have been either approved or are in clinical trials for cancer treatment. The ADC-based treatments of breast cancers, particularly TNBC, MDR and metastatic breast cancers, have shown promise in recent years. This review discusses ADC drug designs, and developed for different types of breast cancer including TNBC, MDR and metastatic breast cancer.

Structure-Activity Relationships of Antibody-Drug Conjugates: A Systematic Review of Chemistry on the Trastuzumab Scaffold.

PMID: 35801843 | Indexed in PubMed

Antibody-drug conjugates (ADCs) are a rapidly growing class of cancer therapeutics that seek to overcome the low therapeutic index of conventional cytotoxic agents. However, realizing this goal has been a significant challenge. ADCs comprise several independently modifiable components, including the antibody, payload, linker, and bioconjugation method. Many approaches have been developed to improve the physical properties, potency, and selectivity of ADCs. The anti-HER-2 antibody trastuzumab, first approved in 1998, has emerged as an exceptional targeting agent for ADCs, as well as a broadly used platform for testing new technologies. The extensive work in this area enables the comparison of various linker strategies, payloads, drug-to-antibody ratios (DAR), and mode of attachment. In this review, these conjugates, ranging from the first clinically approved trastuzumab ADC, ado-trastuzumab emtansine (Kadcyla), to the latest variants are described with the goal of providing a broad overview, as well as enabling the comparison of existing and emerging conjugate technologies.

Antibody-Drug Conjugates for the Treatment of Breast Cancer.

PMID: 34207890 | Indexed in PubMed

Metastatic breast cancer (BC) is currently an incurable disease. Besides endocrine therapy and targeted agents, chemotherapy is often used in the treatment of this disease. However, lack of tumor specificity and toxicity associated with dose exposure limit the manageability of cytotoxic agents. Antibody-drug conjugates (ADCs) are a relatively new class of anticancer drugs. By merging the selectivity of monoclonal antibodies with the cytotoxic properties of chemotherapy, they improve the therapeutic index of antineoplastic agents. Three core components characterize ADCs: the antibody, directed to a target antigen; the payload, typically a cytotoxic agent; a linker, connecting the antibody to the payload. The most studied target antigen is HER2 with some agents, such as trastuzumab deruxtecan, showing activity not only in HER2-positive, but also in HER2-low BC patients, possibly due to a bystander effect.

This property to provide a cytotoxic impact also against off-target cancer cells may overcome the intratumoral heterogeneity of some target antigens. Other cancer-associated antigens represent a strategy for the development of ADCs against triple-negative BC, as shown by the recent approval of sacituzumab govitecan. In this review, we discuss the current landscape of ADC development for the treatment of BC, as well as the possible limitations of this treatment.

Translational PK/PD framework for antibody-drug conjugates to inform drug discovery and development.

PMID: 38738473 | Indexed in PubMed

ADCs represent a transformative class of medicine that combines the specificity of monoclonal antibodies with the potency of highly cytotoxic agents through linkers, aiming to enhance the therapeutic index of cytotoxic drugs. Given the complex molecular structures of ADCs, combining the molecular characteristics of small-molecule drugs and those of large-molecule biotherapeutics, there are several unique considerations when designing nonclinical-to-clinical PK/PD translation strategies. This complexity also demands a thorough understanding of the ADC's components - antibody, linker, and payload - to the overall toxicological, PK/PD, and efficacy profile. ADC development is a multidisciplinary endeavour requiring a strategic integration of nonclinical safety, pharmacology, and PK/PD modelling to translate from bench to bedside successfully. The ADC development underscores the necessity for a robust scientific foundation, leveraging advanced analytical and modelling tools to predict human responses and optimise therapeutic outcomes. This review aims to provide an ADC translational PK/PD framework by discussing unique aspects of ADC nonclinical to clinical PK translation, starting dose determination, and leveraging PK/PD modelling for human efficacious dose prediction and potential safety mitigation.

Antibody-Drug Conjugates-A Tutorial Review.

PMID: 34063364 | Indexed in PubMed

Antibody-drug conjugates (ADCs) are a family of targeted therapeutic agents for the treatment of cancer. ADC development is a rapidly expanding field of research, with over 80 ADCs currently in clinical development and eleven ADCs (nine containing small-molecule payloads and two with biological toxins) approved for use by the FDA. Compared to traditional small-molecule approaches, ADCs offer enhanced targeting of cancer cells along with reduced toxic side effects, making them an attractive prospect in the field of oncology. To this end, this tutorial review aims to serve as a reference material for ADCs and give readers a comprehensive understanding of ADCs; it explores and explains each ADC component (monoclonal antibody, linker moiety and cytotoxic payload) individually, highlights several EMA- and FDA-approved ADCs by way of case studies and offers a brief future perspective on the field of ADC research.

Recent Technological and Intellectual Property Trends in Antibody-Drug Conjugate Research.

PMID: 38399275 | Indexed in PubMed

Antibody-drug conjugate (ADC) therapy, an advanced therapeutic technology comprising antibodies, chemical linkers, and cytotoxic payloads, addresses the limitations of traditional chemotherapy. This study explores key elements of ADC therapy, focusing on antibody development, linker design, and cytotoxic payload delivery. The global rise in cancer incidence has driven increased investment in anticancer agents, resulting in significant growth in the ADC therapy market. Over the past two decades, notable progress has been made, with approvals for 14 ADC treatments targeting various cancers by 2022. Diverse ADC therapies for hematologic malignancies and solid tumors have emerged, with numerous candidates currently undergoing clinical trials. Recent years have seen a noteworthy increase in ADC therapy clinical trials, marked by the initiation of numerous new therapies in 2022. Research and development, coupled with patent applications, have intensified, notably from major companies like Pfizer Inc. (New York, NY, USA), AbbVie Pharmaceuticals Inc. (USA), Regeneron Pharmaceuticals Inc. (Tarrytown, NY, USA), and Seagen Inc. (Bothell, WA, USA). While ADC therapy holds great promise in anticancer treatment, challenges persist, including premature payload release and immune-related side effects. Ongoing research and innovation are crucial for advancing ADC therapy. Future developments may include novel conjugation methods, stable linker designs, efficient payload delivery technologies, and integration with nanotechnology, driving the evolution of ADC therapy in anticancer treatment.

Considerations for the design of antibody drug conjugates (ADCs) for clinical development: lessons learned.

PMID: 38087293 | Indexed in PubMed

Antibody-drug conjugates (ADCs) have emerged as a novel therapeutic strategy that has successfully reached patient treatment in different clinical scenarios. ADCs are formed by an antibody against a specific tumor-associated antigen (TAA), a cytotoxic payload, and a chemical linker that binds both. To this regard, most efforts have been focused on target identification, antibody design and linker optimization, but other relevant aspects for clinical development have not received the necessary attention. In this article using data from approved ADCs, we evaluated all characteristics of these agents, including payload physicochemical properties, in vitro potency, drug antibody ratio (DAR), exposure-response relationships, and clinical development strategies. We suggest that compounds with best options for clinical development include those with optimal payload physicochemical properties and cleavable linkers that would lead to a bystander effect. These modalities can facilitate the development of ADCs in indications with low expression of the TAA. Early clinical development strategies including changes in the schedule of administration with more frequent doses are also discussed in the context of an efficient strategy. In conclusion, we highlight relevant aspects that are needed for the optimal development of ADCs in cancer, proposing options for improvement.

Antibody-drug conjugates for cancer.

PMID: 31478503 | Indexed in PubMed

Antibody-drug conjugates (ADCs) are immunoconjugates comprised of a monoclonal antibody tethered to a cytotoxic drug (known as the payload) via a chemical linker. The ADC is designed to selectively deliver the ultratoxic payload directly to the target cancer cells. To date, five ADCs have received market

approval and over 100 are being investigated in various stages of clinical development. In this Therapeutics paper, we review recent clinical experience with the approved ADCs and other promising late-stage candidates on the horizon, following an overview of the biology and chemistry of ADCs and how the individual components of an ADC (antibody [or target], linker and conjugation chemistry, and cytotoxic payload) influence its activity. We briefly discuss opportunities for enhancing ADC efficacy, drug resistance, and future perspectives for this novel antibody-based molecular platform, which has great potential to make a paradigm shift in cancer chemotherapy.

Antibody-Drug Conjugates in Gynecologic Cancer.

PMID: 37229642 | Indexed in PubMed

The present article reviews the current evidence for antibody-drug conjugates (ADCs) in gynecologic cancer. ADCs consist of a highly selective monoclonal antibody for a tumor-associated antigen and a potent cytotoxic payload conjugated through a linker. Overall, the toxicity profiles of ADCs are manageable. Ocular toxicity is a known class effect of some ADCs and is managed with prophylactic corticosteroid and vasoconstrictor eye drops as well as dose interruptions/holds and dose modifications. In ovarian cancer, mirvetuximab soravtansine, an ADC targeting alpha-folate receptor (FR α), received US Food and Drug Administration (FDA) accelerated approval in November 2022 after data from the single-arm phase III SORAYA trial. A second ADC targeting FR α , STRO-002, received FDA fast track designation in August 2021. Multiple studies with upifitamab rilsodotin, an ADC comprising a NaPi2B-binding antibody, are underway. In cervical cancer, tisotumab vedotin, an ADC-targeting tissue factor, received FDA accelerated approval in September 2021 after the phase II innovaTV 204 trial. Tisotumab vedotin in combination with chemotherapy and other targeted agents is currently being evaluated. Although there are no currently approved ADCs for endometrial cancer, there are many under active evaluation, including mirvetuximab soravtansine. Trastuzumab-deruxtecan (T-DXd), an ADC targeting human epidermal growth factor receptor 2 (HER2), is currently approved for HER2-positive and HER2-low breast cancer and shows promise in endometrial cancer. Like all anticancer treatments, the decision for a patient to undergo therapy with an ADC is a personal choice that balances the potential benefits with the side effects and requires thorough and compassionate support of their physician and care team and shared decision making.

Antibody-drug conjugates: a mini-review. The synopsis of two approved medicines.

PMID: 25625494 | Indexed in PubMed

Targeted drug delivery is a method of delivering bioactive compounds to a patient in a manner that increases the therapeutic index. The main goal of a targeted drug delivery system is to prolong, localize, target and have a protected drug interaction with the diseased tissue. Antibody-drug conjugates (ADC) represent an innovative therapeutic application that combines the unique properties of monoclonal antibodies with the potent cell killing activity of cytotoxic bioactive compounds. ADCs are complex molecules composed of an antibody linked, via a stable, chemical, linker with labile bonds, to a biological active cytotoxic (anticancer) payload or drug. The key components of ADC include a monoclonal antibody, a stable linker and a cytotoxic agent to target a variety of cancers. The present mini-review deals with the examination of clinical use and pharmacological properties, as well as the safety of antibody-drug conjugates that are marketed. Ado-trastuzumab emtansine and brenduximab vedotin were

examined regarding their mechanism of action, pharmacology, clinical use and safety. These ADCs selectively deliver cargoes to tumor cells and provide clinical benefit by minimizing systemic toxicity.
